Lecture 11 (2026-07-19)

Quick Summary

The module transitions from Financial Accounting to Management Accounting. While Financial Accounting is backward-looking and focused on reporting to external parties (using limited public data), Management Accounting is forward-looking and focuses on internal decision-making (using internal, more detailed data). Key focuses in Management Accounting include cost and revenue parameters.

Key Definitions

- **Financial Accounting**: Backward-looking; focuses on reporting; intended for external stakeholders (customers, competitors); relies on limited public domain information.
- **Management Accounting**: Forward-looking; focuses on internal management decisions (e.g., pricing, sourcing materials, staffing models); utilizes comprehensive internal data.

  Direct Cost: Costs directly traceable to a product or process.
  - **Direct Material**: Materials that become part of the finished product (tracked on a per-unit basis when feasible).
  - **Direct Labor**: Labor directly involved in the manufacturing process.
  - **Direct Expense**: Other costs directly related to the production process.
- **Indirect Cost**: Costs that are not directly traceable to the product (e.g., manufacturing/administrative overhead, selling and distribution overhead).
- **Fixed Cost**: Costs that remain constant regardless of production volume or revenue (e.g., rent, salaries).
- **Variable Cost**: Costs that change in direct proportion to production or sales volume (e.g., raw materials).
- **Semi-variable Cost**: Costs containing both a fixed and a variable component (e.g., electricity with a base fee, or a "cap" ride).
- **Opportunity Cost**: The value/return of the next best alternative that is foregone when a choice is made. It represents the "minimum requirement" or "minimum return" a decision must achieve to be justified.
- **Sunk Cost**: Costs that have been already been incurred and cannot be recovered. These are irrelevant for future decision-making. (Example: Old equipment, renovation costs, or a lease that cannot be easily terminated).
- **Marginal Cost**: The incremental cost incurred for every additional unit produced or sold. This is the "relevant cost" for making future decisions.
- **Fixed Asset**: A tangible property owned by the business (e.g., land, buildings, machinery).
- **Relevant Cost**: Costs that are relevant to a specific decision being made "today." Decisions are based on relevant revenue and relevant costs available at the time of the decision.

  Inventory Valuation: Methods for valuing stock, including:
  - **First In, First Out (FIFO)**
  - **Last In, First Out (LIFO)**
  - **Weighted Average Cost**
- **Cash Budget**: A financial plan that estimates cash inflows and outflows of a business to determine available liquidity.
- **Master Budget**: A comprehensive budget including sales, production, direct materials, direct labor, overhead, and cash budgets.

- **Zero-Based Budgeting**: A method where every expense must be justified for each new period. Instead of adjusting previous figures, it starts from zero to ensure resources are allocated to high-value activities.
- **Contribution Margin**: The difference between sales and variable costs.
- **Cost-Volume-Profit (CVP) Analysis**: Analysis used to understand the relationship between cost, volume, and profit to determine how managing quantity, unit costs, and prices affects profit.
- **Contribution Margin per Unit**: Selling price minus variable cost per unit.
- **Contribution Margin Ratio**: Contribution margin divided by sales; represents the proportion of sales that contributes to covering fixed costs and profit.
- **Non-operating Expense**: Expenses not related to core operations (e.g., interest on debentures, discount on debentures).
- **Non-cash Expense**: Expenses that do not involve an actual cash outflow (e.g., depreciation).

Worked Examples

Management Accounting Decision Examples:

- Determining product pricing.
- Deciding between buying raw materials vs. manufacturing them in-house.
- Deciding between hiring permanent staff (Fixed Cost) vs. contract employees (Variable Cost).

Semi-variable Cost Calculation (Cap Ride):

- Base/Fixed: 80 for the first 5 hours.
- Variable: 20 per hour thereafter.

Semi-variable Cost Calculation (Electricity):

- Fixed: A certain amount paid for base usage.
- Variable: Additional charges incurred only when usage exceeds the threshold.

Opportunity Cost (Investment):

- Option A: Bank FD (5% return).
- Option B: Business Investment (12% return).
- **Reasoning:** If you choose the Bank FD, the Opportunity Cost is 12%. The decision is only justified if the FD returns at least 12%.

Opportunity Cost (Career Choice):

- Option A: Full-time course.
- Option B: Stay in job (Salary 150k, potential promotion to 200k).
- **Reasoning:** If you choose the course, your minimum expected salary upon return must be 200k to cover the "cost" of the missed promotion.

Marginal Cost Calculation:

- Scenario: 5 units produced for 50; 6 units produced for 58.
- Calculation: 58 - 50 = 8.
- **Reasoning:** The marginal cost for the 6th unit is 8. This is the "relevant cost" for future decision-making.

Sunk Cost (Software Purchase):

- Scenario: Company spent 40,000 on old software; a new version is available for 100,000.
- Calculation: Sunk Cost = 40,000.
- **Reasoning:** The 40,000 is the sunk cost because it cannot be recovered. The 60,000 difference between the old and new software is a price difference, **not** an opportunity cost.

Cost Classification (Tires, PR, etc.):

- **Tires used for car**: Direct Cost; Variable Cost.
- **Salary of public relations manager**: Indirect Cost; Fixed Cost.
- **Award dinner for suppliers**: Indirect Cost; (Not clearly fixed/variable as it is a one-time expense).
- **Engineer monitoring design**: Indirect Cost (R&D activity).

- **Freight cost**: Direct Cost.
- **Electricity cost**: Variable Cost (or Semi-variable if a base fee exists).
- **Wages for temporary assembly line**: Direct Cost; Variable/Semi-variable.
- **Annual fire insurance**: Fixed Cost (Does not change with production volume; only changes with substantial changes in capacity/value).

Chemical Conversion Calculation:

- Scenario: 1,000 units purchased at 500/unit (Total 500,000).
- Current Condition: Item is unsuitable for current use.
- Options:
  1. Do nothing: Sale price = 0.
  2. Convert: Cost of conversion = 700/unit; Sale price = 700/unit.
  3. Convert (Alternative): Cost of conversion = 600/unit; Sale price = 700/unit.
- **Reasoning:** In all scenarios, the original purchase price (500/unit) is a **Sunk Cost** and must be ignored. It is "already gone." The decision is based only on the **future** cost of conversion vs. the **future** revenue from selling.

Potato Selling Case:

- Scenario: Potatoes purchased for 5,000.
- Context: Expert labor is unavailable; the owner must do the work.
- Comparison: Selling as-is (Price 10,000) vs. making chips (Price 4,000).
- **Reasoning:** Even if the price drops to 4,000, the owner should still sell the items. This is because they are "recovering" part of the 5,000 sunk cost.

Cricket Ticket Comparison (Opportunity Cost):

- Scenario A: 50 free tickets to an orphanage.
- Scenario B: 50 free tickets to college students.
- **Reasoning:** Option B has a higher **Opportunity Cost**. Because college students would have paid for the tickets, giving them away constitutes a "loss of possibility of revenue." If the orphanage audience would not have paid regardless, the opportunity cost of giving them tickets is much lower.

Weighted Average Cost Calculation:

- Scenario: 100 units @ 20, 50 units @ 30, 75 units @ 35.
- Step 1: Calculate total value $(100 \times 20) + (50 \times 30) + (75 \times 35) = 2000 + 1500 + 2625 = 6125$.
- Step 2: Divide by total units $(100 + 50 + 75 = 225)$.
- Calculation: $6125 / 225 \approx 27.33$ (Note: Instructor noted approximately 27.25).
- **Reasoning:** Weighted average cost is always between the lowest and highest costs.

"Happy Days" Case (Contribution Calculation):

- Scenario: 8,000 units sold.
- Revenue: 100,000.
- Variable Cost: 72,000.
- Step 1: Calculate Contribution: $100,000 - 72,000 = 28,000$.
- Step 2: Calculate unit components for decision analysis:
  - Revenue per unit: $100,000 / 8,000$
  - Variable cost per unit: $72,000 / 8,000$
  - Contribution per unit: $28,000 / 8,000$
- **Reasoning:** This method allows management to see how much each additional unit contributes to covering fixed costs and generating profit.

Break-even Calculation (Fixed Cost / Contribution per Unit):

- Scenario: Fixed Cost = 35,000. Contribution Margin per unit = 3.50.
- Calculation: $35,000 / 3.5 = 10,000$ units.
- **Reasoning:** To find how many units are needed to reach a zero profit/loss position, divide the total fixed cost by the amount each unit contributes over its variable cost.

Impact of Quantity Changes on Profit:

- Scenario: If sales increase by 100 units.
- Calculation: $100 \times 3.5 = 350$ increase in net income.
- **Reasoning:** Since fixed costs do not change, every additional unit sold adds exactly the "Contribution per unit" to the net income.

Calculation via Contribution Margin Ratio:

- Scenario: Contribution = 28,000. Sales = 100,000.
- Step 1: Calculate CM Ratio: $28,000 / 100,000 = 28\%$ (0.28).
- Step 2: Calculate profit change from sales volume drop (e.g., 200 units).
- Calculation: $200 \times 0.28 = 56$ (or equivalent logic using volume).
- **Reasoning:** The Calculation using the CM Ratio (200×0.28) and the Calculation using Contribution per unit (200×3.5) result in the same value because the ratio is proportional to the unit margin.

Cash Flow Statement Adjustments:

Depreciation: Add back to Net Profit.

- **Reasoning:** It is a non-cash expense. It was subtracted to calculate Net Profit but did not involve a cash outflow, so it must be added back to find the actual cash flow.

Interest on Debentures / Discount on Debentures: Add back to Net Profit.

- **Reasoning:** These are non-operating (financing) expenses. Since they were deducted from Net Profit but are not part of "operating" activities, they are added back.

Loss on Sale of Assets: Add back to Net Profit.

- **Reasoning:** It is a non-operating item. Because it was a "negative" item that was subtracted to reach net profit, adding it back corrects the cash flow.

Profit on Sale of Assets: Subtract from Net Profit.

- **Reasoning:** It is a non-operating "positive" item that was added to net profit; it must be removed to isolate operating cash flow.

Formulas & Rules

Difference between Direct/Indirect (Financial Accounting focus):

- Direct = Manufacturing/Product related.
- Indirect = Non-production related (Office, Insurance, Selling).

Difference between Fixed/Variable (Management Accounting focus):

- Fixed = Constant regardless of output.
- Variable = Changes in direct proportion to production volume.

- **Opportunity Cost Rule**: It is not a "negative cost." It is the minimum return you must generate from an investment to justify not taking the alternative.

Total vs. Unit Behavior:

- **Total Fixed Cost**: Remains constant (straight line) regardless of volume.
- **Fixed Cost per Unit**: Decreases as volume increases (downward sloping).
- **Total Variable Cost**: Increases in direct proportion to production volume.
- **Variable Cost per Unit**: Remains constant (horizontal line).

Relationship Calculation:

- If you know the cost per unit, you can find the total cost.
- If you know the total cost, you can calculate the variable cost or marginal cost.

Contribution Margin Formula:

- $\text{Contribution} = \text{Sales} - \text{Variable Costs}$

Net Income Calculation (Management Reporting):

- $\text{Net Income} = \text{Contribution} - \text{Fixed Costs}$

Contribution Margin per Unit:

- $\text{CM per Unit} = \text{Selling Price} - \text{Variable Cost per Unit}$

Contribution Margin Ratio:

- $\text{CM Ratio} = \frac{\text{Contribution}}{\text{Sales}}$

Break-even Point (Units):

- $\text{BEP (Units)} = \frac{\text{Fixed Costs}}{\text{Contribution per Unit}}$

Profit from Volume Change:

- $\text{Change in Net Income} = \text{Change in Units} \times \text{Contribution per Unit}$

Alternative Profit from Volume Change (using Ratio):

- $\text{Change in Net Income} = \text{Change in Sales Value} \times \text{Contribution Margin Ratio}$

Tricky Logic & Traps

- **Scope of Accounting**: Do not confuse the goals of Financial vs. Management Accounting. Financial is for **reporting** what happened; Management is for **deciding** what to do next.
- **Direct/Indirect vs. Fixed/Variable**: While they may seem similar, the distinction between Direct/Indirect is primarily the concern of Financial Accounting (production relationship), while Fixed/Variable is the primary concern of Management Accounting (volume relationship).
- **Opportunity Cost Misconception**: It is often mistaken as a "cost" in the sense of an expenditure. It is actually a "lost return." It is used to set a benchmark: if your chosen path does not provide at least the amount of the opportunity cost, the decision is not economically justified.
- **Lease vs. Buy (The "Difference" Trap)**: A student may try to link "Lease vs. Buy" to Opportunity Cost. The instructor clarified that this is a different distinction: Buying results in a **Fixed Asset** (Capital expenditure), while leasing relates to a **Fixed Cost** (Operating expenditure). They are distinct accounting classifications.
- **Fixed Asset vs. Fixed Cost**: Do not confuse these two. A **Fixed Asset** is a piece of property or equipment (e.g., a building or machinery). A **Fixed Cost** is an operational expense (e.g., rent or salary) that does not change with production volume.
- **Sunk Cost Identification**: Sunk costs are items like renovations, old software, or non-recoverable assets. They are irrelevant to future decision-making.
- **Sunk Cost vs. Opportunity Cost**: In a "New vs. Old" comparison (e.g., two types of software), the cost of the old item is the **Sunk Cost**, while the difference in price between the two items is merely a price difference, **not** the Opportunity Cost.
- **Variable Cost per Unit**: Students often think variable costs change per unit; however, the **Variable Cost per Unit** is constant (a horizontal line). Only the **Total** variable cost changes with volume.
- **Insurance as a Fixed Cost**: Even if an insurance premium might change if the building size changes, it is considered a **Fixed Cost** because it does not fluctuate with daily production volumes or output numbers.
- **One-time Expenses**: Items like "Award Dinners" may not strictly fit the "Fixed vs. Variable" binary because they are one-time occurrences rather than costs that fluctuate with volume.
- **Decision Rule (Sunk vs. Relevant)**: "Don't chase what you can't recover." Decisions must be based on "relevant revenue and relevant cost as of today." What was offered or held in the past (Sunk Costs) should not influence current decisions.
- **Opportunity Cost as Revenue Loss**: Opportunity cost is specifically the "loss of possibility of revenue." For example, giving items away to those who would have paid (e.g., college students) results in a higher opportunity cost than giving them to those who would not have paid.
- **Disposal Costs/Negative Value**: If an item is a liability (e.g., hazardous chemicals), it may have a "negative value" or "disposal cost." In such cases, you should act to get rid of it even if the selling price is lower than the cost of disposal, simply to remove the ongoing cost/problem.
- **LIFO and Inflation/Tax Logic**: When prices are rising (inflation), **LIFO** results in a higher Cost of Goods Sold (COGS) and therefore lower profit. Consequently, LIFO is used specifically to **reduce tax liability** during inflation.

- **Function vs. Behavior Distinction**: While the total Net Income remains the same regardless of whether you use a Traditional or Management approach, the **Traditional** approach groups costs by function (e.g., COGS, Administrative), while the **Management** approach groups costs by behavior (e.g., Variable, Fixed).
- **Exclusion of Interest**: In management accounting (strategic decision making), interest is often ignored because it is a **financial cost** rather than a factor related to operational efficiency or manufacturing volume.
- **Proposed Dividend Trap**: Proposed dividends are "financing in nature" and do not impact Net Profit. While some textbooks (specifically Indian ones) include them in calculations to maintain "homing" or consistency with other sections, it is technically and logically incorrect because the dividend hasn't been paid yet. For exam purposes, follow the specific instruction provided by the instructor regarding the presence of this item in the specific calculation you are performing.

  Non-Operating vs. Non-Cash Distinction:
  - **Depreciation** is a **Non-cash, Operating** expense (Add back to Net Profit to find cash flow).
  - **Interest/Discount on Debentures** are **Non-operating, Financing** expenses (Add back because they were subtracted to reach Net Profit but are not operating costs).

  Loss/Profit on Asset Sale:
  - A **Loss** on sale is a non-operating item that was subtracted to reach Net Profit; it must be **added back** to reconcile to cash flow.
  - A **Profit** on sale is a non-operating item that was added to Net Profit; it must be **subtracted** to reconcile to cash flow.
- **The "Sales Decrease" Trap**: When a question describes a "reduction in sales" and asks for the impact on profit, the loss is the **Change in Sales $\times$ Contribution Margin per unit** (or $\times$ CM Ratio). It is **not** the full Selling Price because fixed costs do not change. The calculation should focus only on the portion of the sale that was lost beyond the variable cost.
- **Selection of "Important" Items**: The instructor will not identify specific "important" items for the exam; students are expected to study the entire syllabus.
- **Logic-Based Formulas**: Formulas are not arbitrary; they are based on logic. The exam tests the ability to identify the correct logic to apply to a given scenario.

Exam Pointers

- **Scope of Study**: Study the entire syllabus; the instructor will not provide a list of "important" topics or items.
- **Resource Utilization**: Use the provided PPTs as the primary reference to navigate the logic and select the correct formula for any given calculation.

Sources

- Session 1 - FMA 2026.pdf
- PV ratio practice question.docx
- IFRS (1).pdf
- Make or Buy and Special Order.docx